

Recovery Analysis Report

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Executive summary

The verified simulation completed at day 12. Customer service averaged 100.0%, peak service loss was 0.000, and verified execution cost \$6,069.92. The verifier recorded 1 intervention(s).

Scenario and network

Field	Value
Scenario	A port closure interrupts the main inbound medical-supply route for 5 days.
Affected nodes	M1, D1, R1
Duration	5 days
Severity	70.0%
Network status	Recovering - buffer stock protecting service
Physical imported nodes	7
Data quality	All physical IDs and production capacities preserved

Verifier and negotiation

Field	Value
Decision	APPROVED
Agreement valid	Yes
Negotiation status	verified
Negotiation rounds	1
Invalid executed rate	0.0%

ARDN advisory forecast

Field	Value
Recommended action	Use emergency source
Recovery prediction	11.5 +/- 1.6 days
Predicted service loss	0.035
Predicted cost	\$234.00
Cost interval	\$12.50 - \$455.40
Overflow risk	<1% - Low severe-overflow risk
OOD status	10.61 / threshold 15
Execution status	Within OOD threshold

Strategy comparison

Metric	Rules only	Unverified	Verified
Recovery time	7 days	7 days	7 days
Invalid executed rate	0.0%	0.0%	0.0%
Peak service loss	0.000	0.000	0.000
Average service	100.0%	100.0%	100.0%
Fairness variance	0.0103	0.0937	0.0089
Total cost	\$5,948.44	\$8,099.12	\$6,069.92
Verifier interventions	0 rounds	0 rounds	1 rounds

Interpretation

ARDN is advisory. The verified simulator results are the ground-truth execution outcome, while ARDN values are forecasts with their own definitions and uncertainty.